

Teleparallel Dynamics

First Principles

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From torsion kinematics to field equations

Start with the TEGR torsion tensor $T^\lambda_{\mu\nu} = \partial_\mu e^\lambda_\nu - \partial_\nu e^\lambda_\mu$, superpotential $S^\mu_{\lambda}{}^{\mu\nu}$, and scalar $T = S^\mu_{\lambda}{}^{\mu\nu} T^\lambda_{\mu\nu}$. Vary the action $S = \frac{1}{2\kappa} \int e T d^4x + \int e \mathcal{L}_m d^4x$ with respect to the tetrad e^a_μ . After integrations by parts and index hygiene, the Euler–Lagrange result is:

$$e^a_\lambda \partial_\nu (e S^{\mu\nu}{}_\lambda) - e (S^\nu{}_\rho{}^{\mu\lambda} T_\nu{}^\lambda{}_\rho e^a_\lambda - \frac{1}{2} e^a_\mu T) = \kappa e \Theta^\mu_a \quad (1)$$

1 TEGR Field Equation: Definitions and Properties

Setup and indices

- Spacetime (M, g) with signature $(-, +, +, +)$.
- Greek $\mu, \nu, \rho, \lambda \in \{0, \dots, 3\}$; Latin a, b for Lorentz frame.
- Tetrad e^a_μ , inverse $E_a{}^\mu$, determinant $e = \det(e^a_\mu) = \sqrt{-g}$.
- $\eta_{ab} = \text{diag}(-1, 1, 1, 1)$, $g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu$.

Operators

Weitzenböck connection

$$\Gamma^\rho_{\mu\nu} = E_a{}^\rho (\partial_\nu e^a_\mu + \omega^a_{b\nu} e^b_\mu).$$

Torsion, contortion, superpotential

$$\begin{aligned} T^\rho_{\mu\nu} &:= \Gamma^\rho_{\nu\mu} - \Gamma^\rho_{\mu\nu}, \\ K^\rho_{\mu\nu} &:= \Gamma^\rho_{\mu\nu} - \{\rho_{\mu\nu}\} = \frac{1}{2} (T_\mu{}^\rho{}_\nu + T_\nu{}^\rho{}_\mu - T^\rho{}_{\mu\nu}), \\ S_\rho{}^{\mu\nu} &:= \frac{1}{2} (K^{\mu\nu}{}_\rho + \delta^\mu_\rho T^{\alpha\nu}{}_\alpha - \delta^\nu_\rho T^{\alpha\mu}{}_\alpha). \end{aligned}$$

Scalars and identity

$$T = S_\rho{}^{\mu\nu} T^\rho_{\mu\nu}, \quad B = \frac{2}{e} \partial_\mu (e T^{\mu\nu}{}_\nu), \quad R(g) = -T + B.$$

Variation and field equation

$$S = \frac{1}{2\kappa} \int e T d^4x + \int e \mathcal{L}_m d^4x, \quad \Theta^\mu_a := \frac{1}{e} \frac{\delta(e \mathcal{L}_m)}{\delta e^a_\mu}.$$

Euler–Lagrange equation:

$$e^a{}_\lambda \partial_\nu (e S^{\mu\nu}{}_\lambda) - e (S_\rho{}^{\nu\mu} T_\nu{}^\lambda{}_\rho e^a{}_\lambda - \frac{1}{2} e^a{}_\mu T) = \kappa e \Theta^\mu{}_a. \quad (2)$$

Properties

- Symmetric part $\Rightarrow$ Einstein equations via $R = -T + B$.
- Local Lorentz invariance with inertial $\omega^a{}_{b\mu}$.
- Conservation: $\nabla_\mu T^\mu{}_\nu = 0$; equivalently $\partial_\mu (e \Theta^\mu{}_a) = 0$ on shell.

Dimension ledger (SI, $c = 1$)

$$[T^\rho{}_{\mu\nu}] = \text{m}^{-1}, \quad [S_\rho{}^{\mu\nu}] = \text{m}^{-1}, \quad [T] = \text{m}^{-2}, \quad [\Theta^\mu{}_a] = \text{J m}^{-3}, \quad [\kappa] = \text{m}^2/G.$$

Linearization and GEM map

Let $e^a{}_\mu = \delta^a_\mu + \frac{1}{2} h^a{}_\mu$. Then $T \sim \partial h$, $S \sim \partial h$ and

$$\partial_\nu S^{\mu\nu}{}_\lambda \approx \frac{1}{2} \bar{h}^\mu{}_\lambda \quad \text{in a harmonic-like teleparallel gauge.}$$

Identify $\phi_g := -\frac{1}{2} h_{00}$, $A_{gi} := h_{0i}$ to obtain GEM potentials.

Common gauges

- Weitzenböck: $\omega^a{}_{b\mu} = 0$.
- Time gauge: $e^0{}_i = 0$ for 3+1 split.
- Harmonic teleparallel: $\partial_\nu (e S^{\mu\nu}{}_\lambda) = 0$.

2 Weak-field, gauge fixing, and the GEM/HGEM map

Linearization and gauge

Set $e^a{}_\mu = \delta^a_\mu + \frac{1}{2} h^a{}_\mu$ with $|h| \ll 1$. Define $h_{\mu\nu} := \eta_{a(\mu} h^a{}_{\nu)}$ and trace-reverse $\bar{h}_{\mu\nu} := h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h$, $h := \eta^{\alpha\beta} h_{\alpha\beta}$. Teleparallel harmonic gauge:

$$\partial_\nu (e S^{\mu\nu}{}_\lambda) = 0 \quad \Rightarrow \quad \partial^\nu \bar{h}_{\mu\lambda, \nu} = 0 \quad (\text{to first order}).$$

Then the field equation reduces to the standard wave system

$$\bar{h}_{\mu\nu} = -2\kappa T_{\mu\nu}, \quad := -\frac{1}{c^2} \partial_t^2 + \nabla^2. \quad (3)$$

GEM potentials and fields

Choose potentials (c explicit for units)

$$\Phi_g := -\frac{c^2}{2} h_{00}, \quad \mathbf{A}_g := c^2 (h_{01}, h_{02}, h_{03}),$$

and fields

$$\mathbf{E}_g := -\nabla\Phi_g - \partial_t \mathbf{A}_g, \quad \mathbf{B}_g := \nabla \times \mathbf{A}_g.$$

With $\rho := T^{00}/c^2$ and $\mathbf{J} := (T^{01}, T^{02}, T^{03})/c$, Eq. (3) gives the Maxwell-form system

$$\begin{aligned} \nabla \cdot \mathbf{E}_g &= -4\pi G \rho, \\ \nabla \times \mathbf{B}_g - \frac{1}{c^2} \partial_t \mathbf{E}_g &= -\frac{4\pi G}{c^2} \mathbf{J}, \\ \nabla \cdot \mathbf{B}_g &= 0, \quad \nabla \times \mathbf{E}_g + \partial_t \mathbf{B}_g = 0. \end{aligned}$$

HGEM: harmonic modes

Assume time-harmonic fields $e^{-i\omega t}$. Then $\partial_t \rightarrow -i\omega$ and

$$(\nabla^2 + k^2) \bar{h}_{\mu\nu}(\mathbf{x}) = -2\kappa T_{\mu\nu}(\mathbf{x}), \quad k := \frac{\omega}{c}.$$

Cavity or lattice boundaries yield resonant spectra by the Helmholtz operator. Equivalently for potentials:

$$(\nabla^2 + k^2) \Phi_g = -4\pi G \rho, \quad (\nabla^2 + k^2) \mathbf{A}_g = -\frac{4\pi G}{c^2} \mathbf{J}.$$

Operator map with units

$$\underbrace{\mathcal{E}_{\text{TEGR}}[e]}_{\text{torsion action}} \xrightarrow[\text{lin}]{\text{gauge}} \underbrace{\bar{h}_{\mu\nu} = -2\kappa T_{\mu\nu}}_{\text{wave}} \xrightarrow{\Phi_g, \mathbf{A}_g} \underbrace{\begin{cases} \nabla \cdot \mathbf{E}_g = -4\pi G \rho \\ \nabla \times \mathbf{B}_g - \frac{1}{c^2} \partial_t \mathbf{E}_g = -\frac{4\pi G}{c^2} \mathbf{J} \\ \nabla \cdot \mathbf{B}_g = 0, \quad \nabla \times \mathbf{E}_g + \partial_t \mathbf{B}_g = 0 \end{cases}}_{\text{GEM}} \xrightarrow{e^{-i\omega t}} \underbrace{(\nabla^2 + k^2) \{\Phi_g, \mathbf{A}_g\} = \text{sources}}_{\text{HGEM}}.$$

$$\Phi_g = -\frac{c^2}{2} h_{00}, \quad \mathbf{A}_g = c^2 h_{0i} \Rightarrow \begin{cases} \nabla \cdot \mathbf{E}_g = -4\pi G \rho, \\ \nabla \times \mathbf{B}_g - \frac{1}{c^2} \partial_t \mathbf{E}_g = -\frac{4\pi G}{c^2} \mathbf{J} \end{cases} \quad (4)$$

Units

$$[\Phi_g] = \text{m}^2 \text{s}^{-2}, [\mathbf{A}_g] = \text{m}^2 \text{s}^{-1}, [\mathbf{E}_g] = \text{m} \text{s}^{-2}, [\mathbf{B}_g] = \text{s}^{-1}.$$

3 Identity $R = -T + B$ and field redefinitions

Connection split and tensors

Write the Levi-Civita connection ${}_{\text{LC}}\Gamma_{\mu\nu}^\rho$ and the Weitzenböck one $\Gamma_{\mu\nu}^\rho$. Contortion: $K_{\mu\nu}^\rho := \Gamma_{\mu\nu}^\rho - {}_{\text{LC}}\Gamma_{\mu\nu}^\rho$. Torsion: $T_{\mu\nu}^\rho := \Gamma_{\nu\mu}^\rho - \Gamma_{\mu\nu}^\rho$, trace $T^\mu := T^{\mu\nu}_\nu$. Superpotential:

$$S_\rho{}^{\mu\nu} = \frac{1}{2} (K^{\mu\nu}_\rho + \delta_\rho^\mu T^{\alpha\nu}_\alpha - \delta_\rho^\nu T^{\alpha\mu}_\alpha).$$

Torsion scalar $T := S_\rho{}^{\mu\nu} T_{\mu\nu}^\rho$. Boundary term $B := \frac{2}{e} \partial_\mu (e T^\mu)$.

Derivation of $R = -T + B$

Use ${}_{\text{LC}}\Gamma^\rho_{\mu\nu} = \Gamma^\rho_{\mu\nu} - K^\rho_{\mu\nu}$ and expand the Ricci scalar $R := g^{\mu\nu} R_{\mu\nu}({}_{\text{LC}}\Gamma)$ with $R_{\mu\nu}({}_{\text{LC}}\Gamma) = \partial_\rho {}_{\text{LC}}\Gamma^\rho_{\mu\nu} - \partial_\nu {}_{\text{LC}}\Gamma^\rho_{\mu\rho} + {}_{\text{LC}}\Gamma^\rho_{\sigma\rho} {}_{\text{LC}}\Gamma^\sigma_{\mu\nu} - {}_{\text{LC}}\Gamma^\rho_{\sigma\nu} {}_{\text{LC}}\Gamma^\sigma_{\mu\rho}$. Because the Weitzenböck curvature vanishes, $R(\Gamma) = 0$, the remaining terms are quadratic in K plus a total divergence. After algebra and using the definitions above one obtains

$$R(g) = -T + \frac{2}{e} \partial_\mu (e T^\mu) \equiv -T + B. \quad (5)$$

Action equivalence

TEGR action $S[e] = \frac{1}{2\kappa} \int e T d^4x$ differs from the Einstein–Hilbert action $S_{\text{EH}}[g] = -\frac{1}{2\kappa} \int e R d^4x$ by the surface integral $\frac{1}{2\kappa} \int e B d^4x = \frac{1}{\kappa} \int \partial_\mu (e T^\mu) d^4x$. With standard falloff or fixed boundary data the variational problems are equivalent.

Field redefinitions to GEM variables

Linearize $e^a_\mu = \delta^a_\mu + \frac{1}{2} h^a_\mu$ and set

$$h_{00} = -\frac{2}{c^2} \Phi_g, \quad h_{0i} = \frac{1}{c^2} A_{gi}, \quad h_{ij} = h_{ij}^{\text{TT}} + \partial_{(i} \xi_{j)} + \delta_{ij} \sigma.$$

Define fields

$$\mathbf{E}_g = -\nabla \Phi_g - \partial_t \mathbf{A}_g, \quad \mathbf{B}_g = \nabla \times \mathbf{A}_g.$$

Matching the Newtonian limit and Lense–Thirring term fixes the normalizations above, yielding the Maxwell-form GEM equations quoted previously.

$$h_{00} = -\frac{2}{c^2} \Phi_g, \quad h_{0i} = \frac{1}{c^2} A_{gi} \implies \begin{cases} \nabla \cdot \mathbf{E}_g = -4\pi G \rho, \\ \nabla \times \mathbf{B}_g - \frac{1}{c^2} \partial_t \mathbf{E}_g = -\frac{4\pi G}{c^2} \mathbf{J} \end{cases} \quad (6)$$

Units and scaling check

$[\Phi_g] = \text{m}^2 \text{s}^{-2}$, $[A_g] = \text{m}^2 \text{s}^{-1}$, $[\mathbf{E}_g] = \text{m} \text{s}^{-2}$, $[\mathbf{B}_g] = \text{s}^{-1}$, consistent with (6) and with $R = -T + B$ in (5).

4 Clean gauge-fixed linear map, units, and reference implementation

Lorenz-type gauge and potentials

Adopt the teleparallel Lorenz gauge

$$\partial_\nu (e S^{\mu\nu}{}_\lambda) = 0 \implies \partial_\mu \bar{h}^\mu{}_\lambda = 0$$

to first order. Define

$$\Phi_g := -\frac{c^2}{2} h_{00}, \quad \mathbf{A}_g := c^2 (h_{01}, h_{02}, h_{03}), \quad \mathbf{E}_g = -\nabla \Phi_g - \partial_t \mathbf{A}_g, \quad \mathbf{B}_g = \nabla \times \mathbf{A}_g.$$

Clean map (operator form with units)

Sources $\rho := T^{00}/c^2 [\rho] = \text{kg m}^{-3}$, $\mathbf{J} := (T^{01}, T^{02}, T^{03})/c [\mathbf{J}] = \text{kg m}^{-2} \text{s}^{-1}$. With $\square := -c^{-2}\partial_t^2 + \nabla^2$,

$$\begin{aligned} \Phi_g &= 4\pi G \rho, \\ \mathbf{A}_g &= \frac{4\pi G}{c^2} \mathbf{J}, \\ \nabla \cdot \mathbf{A}_g + \frac{1}{c^2} \partial_t \Phi_g &= 0 \end{aligned} \tag{7}$$

and therefore

$$\nabla \cdot \mathbf{E}_g = -4\pi G \rho, \quad \nabla \times \mathbf{B}_g - \frac{1}{c^2} \partial_t \mathbf{E}_g = -\frac{4\pi G}{c^2} \mathbf{J}, \quad \nabla \cdot \mathbf{B}_g = 0, \quad \nabla \times \mathbf{E}_g + \partial_t \mathbf{B}_g = 0.$$

Retarded solutions (free space)

With $t_r = t - \frac{|\mathbf{x} - \mathbf{x}'|}{c}$,

$$\begin{aligned} \Phi_g(\mathbf{x}, t) &= G \int_{\mathbb{R}^3} \frac{\rho(\mathbf{x}', t_r)}{|\mathbf{x} - \mathbf{x}'|} d^3 x', \\ \mathbf{A}_g(\mathbf{x}, t) &= \frac{G}{c^2} \int_{\mathbb{R}^3} \frac{\mathbf{J}(\mathbf{x}', t_r)}{|\mathbf{x} - \mathbf{x}'|} d^3 x' \end{aligned} \tag{8}$$

HGEM (time-harmonic) reduction

Assume $e^{-i\omega t}$ ($k = \omega/c$):

$$(\nabla^2 + k^2) \Phi_g = 4\pi G \rho_\omega, \quad (\nabla^2 + k^2) \mathbf{A}_g = \frac{4\pi G}{c^2} \mathbf{J}_\omega,$$

with boundary conditions fixing the resonant spectrum k_n .

Unit ledger (SI)

$$\begin{aligned} [\Phi_g] &= \text{m}^2 \text{s}^{-2}, \quad [\mathbf{A}_g] = \text{m}^2 \text{s}^{-1}, \quad [\mathbf{E}_g] = \text{m s}^{-2}, \quad [\mathbf{B}_g] = \text{s}^{-1}, \\ [\rho] &= \text{kg m}^{-3}, \quad [\mathbf{J}] = \text{kg m}^{-2} \text{s}^{-1}, \quad [G] = \text{m}^3 \text{kg}^{-1} \text{s}^{-2}, \quad [c] = \text{m s}^{-1}. \end{aligned}$$

Reference implementation (discrete)

1. Input $\rho(\mathbf{x}, t)$, $\mathbf{J}(\mathbf{x}, t)$ on a grid; choose Lorenz gauge.
2. Solve (7) by
 - time domain: leapfrog for with absorbing boundaries; or
 - frequency domain: Helmholtz solver $(\nabla^2 + k^2)$ with Robin/PEC analog.
3. Recover fields: $\mathbf{E}_g = -\nabla \Phi_g - \partial_t \mathbf{A}_g$, $\mathbf{B}_g = \nabla \times \mathbf{A}_g$.
4. Validate units by checking $\|\nabla \cdot \mathbf{E}_g + 4\pi G \rho\|_2$ and the Lorenz residual.

5 HGEM resonance, energy, and flux

Vacuum wave relations

In vacuum ($\rho = \mathbf{J} = 0$) the GEM system reduces to

$$\nabla \cdot \mathbf{E}_g = 0, \quad \nabla \times \mathbf{B}_g - \frac{1}{c^2} \partial_t \mathbf{E}_g = 0, \quad \nabla \cdot \mathbf{B}_g = 0, \quad \nabla \times \mathbf{E}_g + \partial_t \mathbf{B}_g = 0.$$

Plane wave ansatz $\propto e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$ gives

$$\mathbf{k} \cdot \mathbf{E}_g = 0, \quad \mathbf{k} \cdot \mathbf{B}_g = 0, \quad \omega = c |\mathbf{k}|, \quad \mathbf{B}_g = \frac{1}{\omega} \mathbf{k} \times \mathbf{E}_g = \frac{1}{c} \hat{\mathbf{k}} \times \mathbf{E}_g.$$

Poynting-type balance (sign convention)

Define effective density and flux (choosing the sign so work on matter is positive):

$$u_g := -\frac{1}{8\pi G} \left(\frac{\|\mathbf{E}_g\|^2}{c^2} + \|\mathbf{B}_g\|^2 \right), \quad \mathbf{S}_g := -\frac{1}{4\pi G} \mathbf{E}_g \times \mathbf{B}_g.$$

Then, from the GEM equations,

$$\partial_t u_g + \nabla \cdot \mathbf{S}_g = -\frac{1}{c^2} \mathbf{J} \cdot \mathbf{E}_g, \quad (9)$$

which mirrors electromagnetic energy balance. For a vacuum plane wave, $\|\mathbf{S}_g\|/|u_g| = c$.

Resonant HGEM modes

For time-harmonic fields $\mathbf{E}_g(\mathbf{x}, t) = \Re[\mathbf{E}_\omega(\mathbf{x})e^{-i\omega t}]$, $\mathbf{B}_g(\mathbf{x}, t) = \Re[\mathbf{B}_\omega(\mathbf{x})e^{-i\omega t}]$, the source-free problem reduces to Helmholtz systems

$$(\nabla^2 + k^2) \Phi_{g,\omega} = 0, \quad (\nabla^2 + k^2) \mathbf{A}_{g,\omega} = 0, \quad k = \omega/c,$$

with Lorenz constraint $\nabla \cdot \mathbf{A}_{g,\omega} + i \frac{\omega}{c^2} \Phi_{g,\omega} = 0$. Boundary conditions:

- Dirichlet: $\Phi_{g,\omega}|_{\partial\Omega} = 0$, $\mathbf{n} \times \mathbf{A}_{g,\omega}|_{\partial\Omega} = 0$.
- Neumann/Robin: $\mathbf{n} \cdot \nabla \Phi_{g,\omega} + \alpha \Phi_{g,\omega} = 0$, similarly for $\mathbf{A}_{g,\omega}$.

Eigenfrequencies $\{k_n\}$ follow from these constraints; orthogonality holds with weight 1 for scalars and the standard L^2 inner product for vectors under the gauge constraint.

Mode orthogonality and normalization

Let Φ_m, Φ_n solve $(\nabla^2 + k_m^2) \Phi_m = 0$, $(\nabla^2 + k_n^2) \Phi_n = 0$ with homogeneous boundary data. Then

$$(k_m^2 - k_n^2) \int_{\Omega} \Phi_m \Phi_n dV = \int_{\partial\Omega} (\Phi_m \partial_n \Phi_n - \Phi_n \partial_n \Phi_m) dA.$$

For identical homogeneous conditions the surface term vanishes, yielding $\int_{\Omega} \Phi_m \Phi_n dV = 0$ for $m \neq n$. An analogous result holds for $\mathbf{A}_{g,\omega}$ after projection onto the Lorenz subspace.

Cavity power balance

Integrate (9) over a volume Ω with outward normal $\mathbf{n}$:

$$\frac{d}{dt} \int_{\Omega} u_g dV + \int_{\partial\Omega} \mathbf{S}_g \cdot \mathbf{n} dA = -\frac{1}{c^2} \int_{\Omega} \mathbf{J} \cdot \mathbf{E}_g dV. \quad (10)$$

In lossless, source-free cavities the time average of $\int_{\partial\Omega} \mathbf{S}_g \cdot \mathbf{n} dA$ vanishes, fixing modal Q by boundary impedance or radiation conditions.

6 Observables: force law and Lense–Thirring precession

Geodesic to GEM force law (slow motion)

For a test worldline $x^\mu(\tau)$ with 3-velocity $\mathbf{v}$ and $|\mathbf{v}| \ll c$, expand the geodesic equation with

$$g_{00} = -1 + \frac{2\Phi_g}{c^2}, \quad g_{0i} = \frac{1}{c^2} A_{gi}, \quad g_{ij} = \delta_{ij} + O(h_{ij}),$$

using the GEM map of the previous sections. Keeping terms up to $O(v/c)$ and neglecting h_{ij} -contributions,

$$\mathbf{a} = -c^2 \Gamma_{00}^i \hat{\mathbf{e}}_i - 2c \Gamma_{0j}^i v^j \hat{\mathbf{e}}_i = -\nabla \Phi_g - \partial_t \mathbf{A}_g + \kappa_B \mathbf{v} \times \mathbf{B}_g + O\left(\frac{v^2}{c^2}, \nabla h_{ij}\right),$$

where $\mathbf{E}_g = -\nabla \Phi_g - \partial_t \mathbf{A}_g$, $\mathbf{B}_g = \nabla \times \mathbf{A}_g$. With the normalizations adopted here (Sec. *Clean map*), $\kappa_B = 2$.

$$\mathbf{a} = \mathbf{E}_g + 2 \mathbf{v} \times \mathbf{B}_g \quad (\text{to } O(v/c)) \quad (11)$$

Unit check $[\mathbf{E}_g] = \text{m s}^{-2}$, $[\mathbf{B}_g] = \text{s}^{-1}$, so $[\mathbf{v} \times \mathbf{B}_g] = \text{m s}^{-2}$ as required.

Gyroscope (spin) precession

Let $\mathbf{S}$ be a gyroscope spin, transported via Fermi–Walker to linear order. In GEM variables the precession rate is

$$\frac{d\mathbf{S}}{dt} = \Omega_{\text{LT}} \times \mathbf{S} \quad \text{with} \quad \Omega_{\text{LT}} = \frac{1}{2} \mathbf{B}_g$$

under the present normalization and to leading order in v/c .

$$\Omega_{\text{LT}} = \frac{1}{2} \mathbf{B}_g, \quad \dot{\mathbf{S}} = \Omega_{\text{LT}} \times \mathbf{S} \quad (12)$$

Far-field of a rotating source (rigid, compact)

For mass current density $\mathbf{J} = \rho \mathbf{v}$ with compact support, the retarded potentials (Sec. *Retarded solutions*) give, at distances r large compared to the source size and to leading multipole order,

$$\mathbf{A}_g(\mathbf{x}, t) \simeq \frac{G}{c^2} \frac{\mathbf{J}_{\text{ang}} \times \mathbf{r}}{r^3}, \quad \mathbf{B}_g(\mathbf{x}, t) \simeq \frac{G}{c^2} \frac{3\mathbf{n}(\mathbf{n} \cdot \mathbf{J}_{\text{ang}}) - \mathbf{J}_{\text{ang}}}{r^3},$$

where $\mathbf{J}_{\text{ang}} = \int \mathbf{x}' \times (\rho \mathbf{v}) d^3x'$ is the source angular momentum and $\mathbf{n} = \mathbf{r}/r$.

$$\mathbf{B}_g(\mathbf{x}) \propto \frac{G}{c^2} \frac{3\mathbf{n}(\mathbf{n} \cdot \mathbf{J}_{\text{ang}}) - \mathbf{J}_{\text{ang}}}{r^3}, \quad r \gg \text{source size} \quad (13)$$

Newtonian and frame-dragging checks

Static limit ($\partial_t = 0$, $\mathbf{A}_g = \mathbf{0}$): $\mathbf{a} = -\nabla\Phi_g$ reproduces Newtonian gravity. Stationary rotating source: $\mathbf{a} = \mathbf{E}_g + 2\mathbf{v} \times \mathbf{B}_g$ yields Lense–Thirring frame dragging consistent with (12).

7 Covariant TEGR: inertial spin connection and antisymmetric sector

Local Lorentz covariance

A local Lorentz transformation $\Lambda^a_b(x) \in SO(1, 3)$ acts as $e^a_\mu \rightarrow \Lambda^a_b e^b_\mu$. To keep the Weitzenböck curvature zero and maintain local Lorentz invariance, include the *inertial* spin connection

$$\omega^a_{b\mu} = (\Lambda^{-1})^a_c \partial_\mu \Lambda^c_b, \quad R^a_{b\mu\nu}(\omega) \equiv 0.$$

The teleparallel connection is $\Gamma^\rho_{\mu\nu} = E_a^\rho (\partial_\nu e^a_\mu + \omega^a_{b\nu} e^b_\mu)$.

$$\omega^a_{b\mu} = (\Lambda^{-1})^a_c \partial_\mu \Lambda^c_b \quad (\text{pure gauge, inertial}) \quad (14)$$

Action and variations

TEGR action (matter omitted here):

$$S[e, \omega] = \frac{1}{2\kappa} \int d^4x e T(e, \omega), \quad T = S_\rho^{\mu\nu} T^\rho_{\mu\nu}.$$

T depends on ω only through a total divergence (the B term in $R = -T + B$). Hence the variation w.r.t. ω is a boundary term:

$$\delta_\omega S = \frac{1}{2\kappa} \int d^4x \partial_\mu \left[e \Xi^\mu_{ab}(e, \omega) \delta \omega^{ab}_\nu g^{\nu\mu} \right],$$

so no field equations arise from ω ; it encodes pure inertia.

$$\frac{\delta S}{\delta \omega^a_{b\mu}} \equiv 0 \quad (\text{up to a surface term}) \quad (15)$$

Antisymmetric sector

Varying w.r.t. the tetrad yields the mixed equation (Sec. *TEGR Field Equation*). In the covariant formalism with (14), the antisymmetric part in (μ, λ) vanishes identically, leaving the symmetric Einstein sector equivalent to GR.

$$E_{[\mu\lambda]}(e, \omega) \equiv 0 \quad \Rightarrow \quad E_{(\mu\lambda)} = \kappa T_{\mu\lambda} \quad (16)$$

Pure-tetrad gauge vs covariant form

Setting $\omega^a_{b\mu} = 0$ (*pure tetrad*) breaks manifest local Lorentz invariance. Then $E_{[\mu\lambda]} = 0$ becomes a non-trivial constraint that selects allowed tetrads (“good tetrads”) free of inertial contamination. The covariant prescription (14) restores invariance for any tetrad related by $\Lambda(x)$.

Recipe: extracting ω for a given tetrad

1. Choose $\Lambda^a_b(x)$ that maps the tetrad to a *reference* inertial tetrad $\tilde{e}^a_\mu = \Lambda^a_b e^b_\mu$ with vanishing torsion in the absence of gravity.
2. Compute $\omega^a_{b\mu}$ via (14).
3. Build $\Gamma^\rho_{\mu\nu}$ and the tensors $T^\rho_{\mu\nu}$, $S^\rho{}^{\mu\nu}$.

Boundary consistency

Because $R = -T + B$, the difference between TEGR and Einstein–Hilbert actions is a surface term $\frac{1}{2\kappa} \int d^4x \, e \, B = \frac{1}{\kappa} \int d^4x \, \partial_\mu (e \, T^\mu)$. Fixing either e^a_μ on ∂M or adding the corresponding boundary counterterm gives a well-posed variational problem consistent with (15).

8 Bimetric teleparallel extension and HGEM screening

Setup and interaction

Introduce two tetrads $(e^a_\mu, \tilde{e}^a_\mu)$ with metrics $g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu$ and $f_{\mu\nu} = \eta_{ab} \tilde{e}^a_\mu \tilde{e}^b_\nu$. Each carries an inertial spin connection to ensure local Lorentz covariance. The ghost-free interaction depends on $S^\mu{}_\nu := (\sqrt{g^{-1}f})^\mu{}_\nu$ via

$$\mathcal{L}_{\text{int}} = m^2 \sum_{n=0}^4 \beta_n e_n(S),$$

with e_n the elementary symmetric polynomials. Matter couples to $g_{\mu\nu}$ only.

Linearization and eigenstates

Linearize around Minkowski: $e^a_\mu = \delta^a_\mu + \frac{1}{2} h^a_\mu$, $\tilde{e}^a_\mu = \delta^a_\mu + \frac{1}{2} \ell^a_\mu$, and define metric perturbations $h_{\mu\nu}$, $\ell_{\mu\nu}$. Diagonalize to mass eigenmodes

$$\bar{h}_{\mu\nu}^{(0)} := \cos \vartheta h_{\mu\nu} + \sin \vartheta \ell_{\mu\nu}, \quad \bar{h}_{\mu\nu}^{(m)} := -\sin \vartheta h_{\mu\nu} + \cos \vartheta \ell_{\mu\nu},$$

where ϑ depends on the Planck masses of the two sectors and β_n . The massive mode has Fierz–Pauli mass μ set by m, β_n .

Field equations (Lorenz-like gauge)

With $\partial^\nu \bar{h}_{\mu\nu}^{(A)} = \frac{1}{2} \partial_\mu \bar{h}^{(A)}$, trace-reverse $\bar{h}_{\mu\nu}^{(A)} \rightarrow \hat{h}_{\mu\nu}^{(A)}$, and source $T_{\mu\nu}$ coupled to $g_{\mu\nu}$, the linear equations are

$$\begin{aligned} \hat{h}_{\mu\nu}^{(0)} &= -2\kappa \cos \vartheta T_{\mu\nu}, \\ (-\mu^2) \hat{h}_{\mu\nu}^{(m)} &= -2\kappa \sin \vartheta T_{\mu\nu}. \end{aligned}$$

Map each sector to GEM potentials with the same normalizations as before.

$$\begin{aligned} \Phi_0 &= 4\pi G \cos \vartheta \rho, & \mathbf{A}_0 &= \frac{4\pi G}{c^2} \cos \vartheta \mathbf{J}, \\ (\nabla^2 - \mu^2)\Phi_m &= 4\pi G \sin \vartheta \rho, & (\nabla^2 - \mu^2)\mathbf{A}_m &= \frac{4\pi G}{c^2} \sin \vartheta \mathbf{J}. \end{aligned} \tag{17}$$

Observables and screening

Static, spherically symmetric source M gives

$$\Phi_g(r) = -\frac{GM}{r} \cos \vartheta - \frac{GM}{r} \sin \vartheta e^{-\mu r},$$

so the massive admixture yields Yukawa screening at $r \gtrsim \mu^{-1}$. Frame-dragging has the same split with $\mathbf{A}_{0,m}$.

HGEM resonance with mass

Time-harmonic reduction in a cavity Ω :

$$(\nabla^2 + k^2)\{\Phi_0, \mathbf{A}_0\} = \text{sources}, \quad (\nabla^2 + k_m^2)\{\Phi_m, \mathbf{A}_m\} = \text{sources},$$

with $k = \omega/c$ and $k_m^2 = k^2 - \mu^2$. Massive modes resonate only for $\omega > \mu c$; below threshold they are evanescent.

$$k_m^2 = \frac{\omega^2}{c^2} - \mu^2 \quad \Rightarrow \quad \text{propagating if } \omega > \mu c, \text{ evanescent otherwise.} \tag{18}$$

Units

$[\mu] = \text{m}^{-1}$, $[\omega] = \text{s}^{-1}$, $[k] = \text{m}^{-1}$. The mixing angle ϑ is dimensionless; $\cos^2 \vartheta + \sin^2 \vartheta = 1$.

9 Numerical HGEM: constrained Helmholtz formulation

Frequency-domain PDEs

Time-harmonic fields $e^{-i\omega t}$ with $k = \omega/c$:

$$(\nabla^2 + k^2)\Phi_g = 4\pi G \rho_\omega, \quad (\nabla^2 + k^2)\mathbf{A}_g = \frac{4\pi G}{c^2} \mathbf{J}_\omega,$$

with Lorenz constraint $\nabla \cdot \mathbf{A}_g + i \frac{\omega}{c^2} \Phi_g = 0$.

Mixed weak form (Lorenz enforced)

Let $\Omega \subset \mathbb{R}^3$ with boundary Γ and outward normal $\mathbf{n}$. Find $(\Phi, \mathbf{A}, \lambda) \in H^1(\Omega) \times H(\text{curl}, \Omega) \times H^1(\Omega)$ such that for all tests $(\psi, \mathbf{v}, \mu)$:

$$\begin{aligned}
& \int_{\Omega} (\nabla \Phi \cdot \nabla \psi - k^2 \Phi \psi) dV + i \frac{\omega}{c^2} \int_{\Omega} \lambda \psi dV - \int_{\Gamma} \psi \partial_n \Phi dA = 4\pi G \int_{\Omega} \rho_{\omega} \psi dV, \\
& \int_{\Omega} (\nabla \times \mathbf{A}) \cdot (\nabla \times \mathbf{v}) dV - k^2 \int_{\Omega} \mathbf{A} \cdot \mathbf{v} dV + \int_{\Omega} \lambda \nabla \cdot \mathbf{v} dV - \int_{\Gamma} (\mathbf{n} \times \mathbf{v}) \cdot (\nabla \times \mathbf{A}) dA = \frac{4\pi G}{c^2} \int_{\Omega} \mathbf{J}_{\omega} \cdot \mathbf{v} dV, \\
& \int_{\Omega} \mu \nabla \cdot \mathbf{A} dV + i \frac{\omega}{c^2} \int_{\Omega} \mu \Phi dV = 0.
\end{aligned}
\tag{19}$$

Radiation/impedance boundary data

Scalar Sommerfeld on $\Gamma_s \subset \Gamma$:

$$\partial_n \Phi - ik \Phi = 0.$$

Vector Silver–Müller on Γ_v :

$$\mathbf{n} \times (\nabla \times \mathbf{A}) - ik \mathbf{n} \times (\mathbf{n} \times \mathbf{A}) = \mathbf{0}.$$

Impose these by replacing the boundary terms in (19) with

$$\int_{\Gamma_s} ik \psi \Phi dA, \quad \int_{\Gamma_v} ik (\mathbf{n} \times \mathbf{v}) \cdot (\mathbf{n} \times \mathbf{A}) dA.$$

Penalty alternative (no multiplier)

Replace the last line of (19) by a penalty term with $\alpha > 0$:

$$\int_{\Omega} (\nabla \times \mathbf{A} \cdot \nabla \times \mathbf{v} - k^2 \mathbf{A} \cdot \mathbf{v}) dV + \frac{1}{\alpha} \int_{\Omega} (\nabla \cdot \mathbf{A} + i \frac{\omega}{c^2} \Phi) (\nabla \cdot \mathbf{v} + i \frac{\omega}{c^2} \psi) dV = \frac{4\pi G}{c^2} \int_{\Omega} \mathbf{J}_{\omega} \cdot \mathbf{v} dV.$$

Discretization recipe

1. Mesh Ω . Use H^1 Lagrange elements for Φ, λ , Nédélec $H(\text{curl})$ elements for $\mathbf{A}$.
2. Choose boundary partitions Γ_s, Γ_v ; apply Sommerfeld/Silver–Müller.
3. Assemble (19) or the penalty system. Scale rows to keep units balanced.
4. Solve the complex linear system. Recover $\mathbf{E}_g = -\nabla \Phi$, $\mathbf{B}_g = \nabla \times \mathbf{A}$ for monochromatic fields.

Units and conditioning

All volume bilinears are in SI with $[\Phi] = \text{m}^2 \text{s}^{-2}$, $[\mathbf{A}] = \text{m}^2 \text{s}^{-1}$, $[k] = \text{m}^{-1}$. Scale unknowns by c if needed to condition $k \rightarrow 0$ and high- k regimes.

10 Analytic HGEM exemplars: sources and cavities

Static point mass

Solve $\nabla^2 \Phi_g = 4\pi G \rho$ with $\rho(\mathbf{x}) = M \delta^3(\mathbf{x})$. By Green's function in $\mathbb{R}^3$,

$$\Phi_g(r) = -\frac{GM}{r}, \quad \mathbf{E}_g = -\nabla \Phi_g = -\frac{GM}{r^2} \hat{\mathbf{r}}.$$

$$\Phi_g(r) = -GM/r, \quad \mathbf{E}_g = -GM \hat{\mathbf{r}}/r^2 \quad (20)$$

Rigidly rotating uniform sphere (far field)

Mass current $\mathbf{J} = \rho(\boldsymbol{\Omega} \times \mathbf{x}')$, angular momentum $\mathbf{J}_{\text{ang}} = \frac{2}{5}MR^2\boldsymbol{\Omega}$. Retarded potential to dipole order:

$$\mathbf{A}_g(\mathbf{x}) \simeq \frac{G}{c^2} \frac{\mathbf{J}_{\text{ang}} \times \mathbf{r}}{r^3}, \quad \mathbf{B}_g = \nabla \times \mathbf{A}_g \simeq \frac{G}{c^2} \frac{3\mathbf{n}(\mathbf{n} \cdot \mathbf{J}_{\text{ang}}) - \mathbf{J}_{\text{ang}}}{r^3}.$$

$$\mathbf{B}_g(\mathbf{x}) \simeq \frac{G}{c^2} \frac{3\mathbf{n}(\mathbf{n} \cdot \mathbf{J}_{\text{ang}}) - \mathbf{J}_{\text{ang}}}{r^3}, \quad r \gg R \quad (21)$$

Spherical cavity eigenmodes (massless sector)

Time-harmonic fields, $e^{-i\omega t}$, $k = \omega/c$. Scalar potential (TM-like): $\Phi_{g,\ell m}(r, \theta, \phi) = A_{\ell m} j_\ell(kr) Y_{\ell m}$. Dirichlet boundary $\Phi_g|_{r=R} = 0$ gives $j_\ell(kR) = 0$.

$$k_{\ell n} = \frac{x_{\ell n}}{R} \quad (x_{\ell n} \text{ } n\text{-th zero of } j_\ell).$$

Vector potential (TE-like toroidal) expanded in vector spherical harmonics yields the tangential boundary $\mathbf{n} \times \mathbf{A}_g|_{r=R} = 0$ which reduces to $j'_\ell(kR) = 0$ for each (ℓ, m) .

$$k_{\ell n}^{\text{TM}} R = x_{\ell n}, \quad k_{\ell n}^{\text{TE}} R = x'_{\ell n} \quad (22)$$

Cylindrical waveguide (radius a)

Assume $e^{-i\omega t + ik_z z}$ and separation in (r, ϕ) . For TM-like Φ_g with $\Phi_g|_{r=a} = 0$: $J_m(\alpha_{mn}r/a)$, $J_m(\alpha_{mn}) = 0$. For TE-like $\mathbf{A}_g$ with A_ϕ, A_z tangential zero: $J'_m(\alpha'_{mn}) = 0$. Transverse relation $k_\perp^2 = \alpha^2/a^2$ and dispersion $k_\perp^2 + k_z^2 = k^2$.

$$\omega_{\text{cut}} = c \sqrt{k_z^2 + \alpha^2/a^2}, \quad \alpha = \begin{cases} \alpha_{mn} : J_m(\alpha_{mn}) = 0 & \text{(TM)} \\ \alpha'_{mn} : J'_m(\alpha'_{mn}) = 0 & \text{(TE)} \end{cases} \quad (23)$$

Massive (bimetric) correction

Replace $k^2 \rightarrow k^2 - \mu^2$ for the massive mode (Sec. 18): propagation requires $\omega > \sqrt{(ck_z)^2 + c^2\alpha^2/a^2 + \mu^2 c^2}$ in the guide and $k_{\ell n}^2 \rightarrow k_{\ell n}^2 - \mu^2$ in spheres; below threshold modes are evanescent.

$$\text{propagating iff } \omega > \sqrt{(ck_z)^2 + c^2\alpha^2/a^2 + \mu^2 c^2} \quad (24)$$

Units

$[R] = [a] = \text{m}$, $[k] = [k_\perp] = [\alpha]/a = \text{m}^{-1}$, $[\omega] = \text{s}^{-1}$, $[\mathbf{A}_g] = \text{m}^2 \text{s}^{-1}$, $[\mathbf{B}_g] = \text{s}^{-1}$.

11 Characteristics, dispersion, and causality

Principal symbol and hyperbolicity

In the Lorenz gauge the linearized equations are $\bar{h}_{\mu\nu} = -2\kappa T_{\mu\nu}$. For co-vector ξ_α , the principal symbol is

$$\mathcal{P}(\xi) \bar{h}_{\mu\nu} := \eta^{\alpha\beta} \xi_\alpha \xi_\beta \bar{h}_{\mu\nu}.$$

Hyperbolicity follows from $\eta^{\alpha\beta} \xi_\alpha \xi_\beta = 0$ defining the null cone.

$$\mathcal{P}(\xi) = \eta^{\alpha\beta} \xi_\alpha \xi_\beta \Rightarrow \text{char. surface } \eta^{\alpha\beta} \partial_\alpha \Phi \partial_\beta \Phi = 0 \quad (25)$$

Massless sector (TEGR/GR) dispersion

Plane waves $e^{i(\mathbf{k}\cdot\mathbf{x} - \omega t)}$ satisfy

$$-\frac{\omega^2}{c^2} + k^2 = 0 \Rightarrow \omega = ck, \quad v_p = \frac{\omega}{k} = c, \quad v_g = \frac{d\omega}{dk} = c.$$

$$\omega^2 = c^2 k^2 \Rightarrow v_p = v_g = c \quad (26)$$

Massive (bimetric) sector dispersion

From $(-\mu^2) \hat{h}_{\mu\nu}^{(m)} = 0$:

$$-\frac{\omega^2}{c^2} + k^2 - \mu^2 = 0 \Rightarrow \omega = c\sqrt{k^2 + \mu^2}, \quad v_g = \frac{d\omega}{dk} = \frac{c^2 k}{\omega} < c.$$

$$\omega^2 = c^2(k^2 + \mu^2), \quad v_g = \frac{c}{\sqrt{1 + \mu^2/k^2}} (< c) \quad (27)$$

Retarded support and continuity

Retarded solutions (§ Clean map) obey

$$\Phi_g(\mathbf{x}, t) = G \int \frac{\rho(\mathbf{x}', t - |\mathbf{x} - \mathbf{x}'|/c)}{|\mathbf{x} - \mathbf{x}'|} d^3x',$$

so support lies inside the future light cone. Constraint propagation: if $\partial_t \rho + \nabla \cdot \mathbf{J} = 0$ and $\nabla \cdot \mathbf{A}_g + \frac{1}{c^2} \partial_t \Phi_g = 0$ at $t = 0$, then the gauge constraint holds for all t .

$$\partial_t \rho + \nabla \cdot \mathbf{J} = 0 \ \& \ (\nabla \cdot \mathbf{A}_g + \frac{1}{c^2} \partial_t \Phi_g)|_{t=0} = 0 \Rightarrow \nabla \cdot \mathbf{A}_g + \frac{1}{c^2} \partial_t \Phi_g \equiv 0 \quad (28)$$

Causality cones

Massless GEM fields propagate on the null cone of $g_{\mu\nu}$. Massive admixtures propagate inside the null cone with timelike characteristics.

$$\text{massless: } \Delta t = |\Delta \mathbf{x}|/c ; \quad \text{massive: } \Delta t > |\Delta \mathbf{x}|/c$$

(29)

Units

$[\omega] = \text{s}^{-1}$, $[k] = [\mu] = \text{m}^{-1}$, $[v_p] = [v_g] = \text{m s}^{-1}$, consistent with (26)–(27).

12 Variational boundary term and well-posedness

Surface term from the TEGR variation

Vary the TEGR action $S[e] = \frac{1}{2\kappa} \int_M d^4x e T$. After integrations by parts the bulk Euler–Lagrange term yields the field equation (Sec. *TEGR Field Equation*), plus a total divergence

$$\delta S = \frac{1}{2\kappa} \int_M d^4x \partial_\mu \left(e \Xi^\mu_a(e, \partial e) \delta e^a_\nu g^{\nu\lambda} \right) + (\text{bulk}),$$

so a boundary contribution remains when $\delta e^a_\mu \neq 0$ on ∂M .

TEGR $\leftrightarrow$ EH and the B -term

Using $R = -T + B$ with $B = \frac{2}{e} \partial_\mu (e T^\mu)$,

$$S_{\text{EH}}[g] = -\frac{1}{2\kappa} \int_M e R = S[e] - \frac{1}{2\kappa} \int_M e B = S[e] - \frac{1}{\kappa} \oint_{\partial M} d^3\Sigma e T^\mu n_\mu,$$

where n_μ is the outward unit normal and $d^3\Sigma$ the induced measure.

Boundary counterterm for a well-posed problem

To fix e^a_μ on ∂M one cancels the surface variation by adding the teleparallel boundary term

$$S_{\partial M}[e] = -\frac{1}{\kappa} \oint_{\partial M} d^3\Sigma e T^\mu n_\mu.$$

Then $\delta(S + S_{\partial M})$ has no leftover boundary piece for Dirichlet data on e^a_μ .

$$S_{\text{TEGR}}^{\text{well-posed}} = \frac{1}{2\kappa} \int_M d^4x e T - \frac{1}{\kappa} \oint_{\partial M} d^3\Sigma e T^\mu n_\mu \quad (30)$$

Units and orientation

$[e T^\mu n_\mu] = \text{m}^{-2}$, $[d^3\Sigma] = \text{m}^3$ so $[S_{\partial M}] = \text{m}$ times $1/[\kappa] = \text{m}^{-2}$ gives an action with units of $\text{J} \cdot \text{s}$. Choose n_μ pointing outward and adopt the induced tetrad on ∂M to keep the sign in (30) consistent with the $R = -T + B$ identity.

Comparison to GHY

The Einstein–Hilbert action attains a well-posed Dirichlet problem by the Gibbons–Hawking–York term $\frac{1}{\kappa} \oint_{\partial M} \sqrt{|h|} K$. Equation (30) is the teleparallel analogue: both differ by a total divergence yet define equivalent dynamics when the corresponding boundary terms are used.

13 Conserved currents and charges in TEGR

13.1 Superpotential and field equations in Lorentz frame

The fundamental geometric object in TEGR is the superpotential $S_\lambda^{\mu\nu}$, defined in terms of the torsion tensor and contortion. Projecting to a Lorentz frame via the tetrad field e^a_λ yields:

$$S_a^{\mu\nu} := e_a^\lambda S_\lambda^{\mu\nu}, \quad \text{with} \quad S_\lambda^{\mu\nu} = -S_\lambda^{\nu\mu}. \quad (31)$$

The TEGR field equations (derived from the teleparallel Lagrangian) can be expressed in compact form as:

$$\partial_\nu (e S_a^{\mu\nu}) = \frac{\kappa}{2} e (\Theta^\mu_a + t^\mu_a), \quad (32)$$

where $e = \det(e^a_\mu)$ is the tetrad determinant, Θ^μ_a is the matter energy-momentum tensor, and t^μ_a represents the gravitational energy-momentum pseudotensor. The explicit form of t^μ_a , while derivable from the gravitational sector, is not required for the conservation laws presented below.

13.2 Conservation law and total four-momentum

Taking the divergence ∂_μ of both sides of the field equation and exploiting the antisymmetry $S_a^{\mu\nu} = -S_a^{\nu\mu}$ yields the continuity equation:

$$\partial_\mu [e (\Theta^\mu_a + t^\mu_a)] = 0 \quad (33)$$

This establishes the total current $e(\Theta^\mu_a + t^\mu_a)$ as conserved. For any spacelike hypersurface Σ , the associated total four-momentum is therefore conserved.

13.3 Quasilocal charge via Gauss' law

Applying Gauss' theorem to (33) over a spacetime region bounded by two spacelike hypersurfaces Σ_1, Σ_2 and a timelike boundary at infinity, we obtain the quasilocal charge. Using the field equation to express the current in terms of the superpotential yields:

$$P_a(\Sigma) = \frac{1}{\kappa} \int_{\partial\Sigma} e S_a^{0i} dS_i \quad (34)$$

For asymptotically flat spacetimes, this definition reproduces the ADM energy for P_0 and the ADM momentum for P_i .

13.4 Flux balance law

Consider a compact 3-volume $\mathcal{V} \subset \Sigma$ with boundary $\partial\mathcal{V}$ and unit outward normal n_i . The time evolution of the enclosed charge follows:

$$\frac{d}{dt} \int_{\mathcal{V}} e (\Theta^0_a + t^0_a) d^3x = - \int_{\partial\mathcal{V}} e (\Theta^i_a + t^i_a) n_i dA. \quad (35)$$

In vacuum ($\Theta^\mu_a = 0$), this reduces to a purely gravitational energy-momentum flux expressible through the superpotential S_a^{0i} .

13.5 Linearized limit and gravitoelectromagnetism

To first order in metric perturbations around Minkowski spacetime, with $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, the superpotential components yield:

$$e S_0^{0i} \approx -\frac{1}{4} \partial_i h_{00} = \frac{1}{2} \partial_i \Phi_g, \quad (36)$$

$$e S_j^{0i} \approx \frac{1}{2} \epsilon^i_{jk} (\nabla \times \mathbf{A}_g)^k, \quad (37)$$

where $\Phi_g = -h_{00}/2$ and $A_g^i = -h_{0i}/2$ are the gravitoelectromagnetic potentials. Substituting into (34) reproduces the Newtonian energy from Φ_g and the Lense–Thirring momentum from $\mathbf{B}_g = \nabla \times \mathbf{A}_g$, up to the factor κ^{-1} .

13.6 Dimensional analysis

In natural units ($c = 1$): $[S_a^{0i}] = L^{-1}$, $[e] = 1$, $[dS_i] = L^2$, and $[\kappa] = 8\pi G = L^2$. Thus:

$$[P_a] = \frac{[e][S_a^{0i}][dS_i]}{[\kappa]} = \frac{L^{-1} \cdot L^2}{L^2} = L^{-1}. \quad (38)$$

Restoring c : $[\kappa] = [8\pi G/c^3] = \text{TM}^{-1}$, giving $[P_0] = \text{J}$ and $[P_i] = \text{kg m s}^{-1}$ as required for energy and momentum.

Total four-momentum: $P_a = \frac{1}{\kappa} \oint_{\partial\Sigma} e S_a^{0i} dS_i$

(39)

This expression provides a geometrically meaningful and gauge-invariant definition of energy-momentum in teleparallel gravity.

14 Symbols, types, and dimensional audit

14.1 Index conventions and fundamental objects

- **Indices:** Coordinate indices $\mu, \nu, \rho, \lambda, \sigma \in \{0, 1, 2, 3\}$; Lorentz frame indices $a, b, c, d \in \{0, 1, 2, 3\}$; spatial indices $i, j, k \in \{1, 2, 3\}$.
- **Metric signatures:** Minkowski metric $\eta_{ab} = \text{diag}(-1, 1, 1, 1)$; spacetime metric $g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu$.
- **Tetrad formalism:** Tetrad e^a_μ (inverse E_a^μ satisfies $e^a_\mu E_a^\nu = \delta_\mu^\nu$, $e^a_\mu E_b^\mu = \delta_b^a$); determinant $e = \det(e^a_\mu) = \sqrt{-g}$.

14.2 Geometric objects and differential operators

Connections and curvature

- **Spin connection:** $\omega^a_{b\mu} = (\Lambda^{-1})^a_c \partial_\mu \Lambda^c_b$, where $\Lambda^a_b \in \text{SO}^+(1, 3)$ represents local Lorentz transformations.
- **Teleparallel connection:** $\Gamma^\rho_{\mu\nu} = E_a^\rho (\partial_\nu e^a_\mu + \omega^a_{b\nu} e^b_\mu)$; metric-compatible ($\nabla_\mu g_{\nu\rho} = 0$) but has torsion.
- **Levi-Civita connection:** ${}_{\text{LC}}\Gamma^\rho_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu})$; curvature-free in teleparallel geometry.

Torsion and related tensors

- **Torsion tensor:** $T^\rho_{\mu\nu} = \Gamma^\rho_{\nu\mu} - \Gamma^\rho_{\mu\nu} = E_a{}^\rho (\partial_\mu e^a_\nu - \partial_\nu e^a_\mu + \omega^a_{b\mu} e^b_\nu - \omega^a_{b\nu} e^b_\mu)$.
- **Contortion tensor:** $K^\rho_{\mu\nu} = \Gamma^\rho_{\mu\nu} - {}_{\text{LC}}\Gamma^\rho_{\mu\nu} = \frac{1}{2}(T^\rho_{\mu}{}^\nu + T^\rho_{\nu}{}^\mu - T^\rho_{\mu\nu})$.
- **Superpotential:** $S_\rho{}^{\mu\nu} = \frac{1}{2}(K^{\mu\nu}{}_\rho + \delta^\mu_\rho T^{\alpha\nu}{}_\alpha - \delta^\nu_\rho T^{\alpha\mu}{}_\alpha)$.

Scalar invariants and operators

- **Torsion scalar:** $T = S_\rho{}^{\mu\nu} T^\rho_{\mu\nu}$.
- **Boundary term:** $B = \frac{2}{e} \partial_\mu (e T^\mu)$, where $T^\mu = T^{\nu\mu}{}_\nu$.
- **Ricci scalar equivalence:** $R = -T + B$, where R is the Levi-Civita Ricci scalar.
- **D'Alembertian:** $\square = g^{\mu\nu} \nabla_\mu \nabla_\nu = -c^{-2} \partial_t^2 + \nabla^2 + \text{curvature terms}$.

14.3 Matter coupling and gravitoelectromagnetism

Matter energy-momentum

- **Matter Lagrangian:** $\mathcal{L}_m$; variational derivative yields: $\Theta^\mu_a = \frac{1}{e} \frac{\delta(e\mathcal{L}_m)}{\delta e^a_\mu}$.
- **Stress-energy tensor:** $T_{\mu\nu} = E_\nu{}^a \Theta_{\mu a} = -\frac{2}{\sqrt{-g}} \frac{\delta(\sqrt{-g}\mathcal{L}_m)}{\delta g^{\mu\nu}}$.
- **Coupling constant:** $\kappa = \frac{8\pi G}{c^4}$.

Gravitoelectromagnetic potentials and fields

Linearized gravity around Minkowski background $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$:

- **Potentials:** $\Phi_g = -\frac{c^2}{2} h_{00}$, $\mathbf{A}_g = \frac{c^2}{2} (h_{01}, h_{02}, h_{03})$.
- **Fields:** $\mathbf{E}_g = -\nabla \Phi_g - \partial_t \mathbf{A}_g$, $\mathbf{B}_g = \nabla \times \mathbf{A}_g$.
- **Sources:** Mass density $\rho = T^{00}/c^2$, mass current $\mathbf{J}^i = T^{0i}/c$.

14.4 Dimensional analysis in SI units

Fundamental constants:

- $[G] = \text{m}^3 \text{kg}^{-1} \text{s}^{-2}$, $[c] = \text{m s}^{-1}$, $[\hbar] = \text{J} \cdot \text{s}$.
- $[\kappa] = [8\pi G/c^4] = \text{s}^2 \text{kg}^{-1} \text{m}^{-1}$.

Geometric quantities:

- $[e^a_\mu] = 1$ (dimensionless), $[g_{\mu\nu}] = 1$, $[e] = 1$.
- $[\Gamma^\rho_{\mu\nu}] = \text{m}^{-1}$, $[T^\rho_{\mu\nu}] = \text{m}^{-1}$, $[S_\rho{}^{\mu\nu}] = \text{m}^{-1}$.

- $[T] = \text{m}^{-2}, [R] = \text{m}^{-2}$.

Matter and GEM fields:

- $[\Theta^\mu_a] = \text{J m}^{-3} = \text{kg m}^{-1} \text{s}^{-2}, [T_{\mu\nu}] = \text{J m}^{-3}$.
- $[\Phi_g] = \text{m}^2 \text{s}^{-2}, [\mathbf{A}_g] = \text{m}^2 \text{s}^{-1}, [\mathbf{E}_g] = \text{m s}^{-2}, [\mathbf{B}_g] = \text{s}^{-1}$.
- $[\rho] = \text{kg m}^{-3}, [\mathbf{J}] = \text{kg m}^{-2} \text{s}^{-1}$.

14.5 Geometrized units and conversions

Set $G = c = 1$:

- **Length–mass–time equivalence:** $1 \text{ kg} \hat{=} 7.426 \times 10^{-28} \text{ m} \hat{=} 2.477 \times 10^{-36} \text{ s}$.
- **Potentials:** $\Phi_g^{(\text{geo})} = \Phi_g/c^2$ (dimensionless); $\mathbf{A}_g^{(\text{geo})} = \mathbf{A}_g/c$ (dimensionless).
- **Stress–energy:** $T_{\mu\nu}^{(\text{geo})} = (G/c^4) T_{\mu\nu}^{(\text{SI})}; [\rho^{(\text{geo})}] = \text{m}^{-2}$.

14.6 Consistency checks via field equations

Linearized field equations:

$$\bar{h}_{\mu\nu} = -2\kappa T_{\mu\nu} \Rightarrow [\kappa T_{\mu\nu}] = [\partial^2 h_{\mu\nu}] = \text{m}^{-2}$$

GEM analogies:

$$\nabla \cdot \mathbf{E}_g = -4\pi G \rho \Rightarrow [\mathbf{E}_g]/[L] = [G\rho]$$

$$\nabla \times \mathbf{B}_g = -\frac{4\pi G}{c^2} \mathbf{J} + \frac{1}{c^2} \partial_t \mathbf{E}_g \Rightarrow [\mathbf{B}_g]/[L] = [G\mathbf{J}]/c^2$$

(40)

14.7 Mathematical type signatures

- **Tetrads:** $e^a_\mu \in \Gamma(T^*M \otimes \mathbb{R}^{1,3}), E_a^\mu \in \Gamma(TM \otimes (\mathbb{R}^{1,3})^*)$.
- **Torsion:** $T^\rho_{\mu\nu} \in \Gamma(TM \otimes \Lambda^2 T^*M)$.
- **Superpotential:** $S_\rho^{\mu\nu} \in \Gamma(T^*M \otimes \Lambda^2 TM)$.
- **GEM fields:** $\Phi_g \in C^\infty(M), \mathbf{A}_g \in \Gamma(T^*\Sigma)$ (on spatial slice Σ), $\mathbf{E}_g, \mathbf{B}_g \in \Gamma(T\Sigma)$.

Key dimensional relations:

$$[P_a] = \frac{[e][S_a^{0i}][dS_i]}{[\kappa]} = \text{J (energy), kg} \cdot \text{m} \cdot \text{s}^{-1} \text{ (momentum)}$$

$$[\text{Torsion}] = \text{m}^{-1}, [\text{Torsion scalar}] = \text{m}^{-2}$$

$$[\text{Gravitational action}] = [\kappa^{-1} \int T e d^4x] = \text{J} \cdot \text{s}$$

(41)

15 Post-Newtonian corrections in TEGR

15.1 Expansion scheme and gauge conditions

The post-Newtonian expansion is formulated in terms of the small dimensionless parameter:

$$\epsilon \sim \frac{v^2}{c^2} \sim \frac{U}{c^2} \ll 1,$$

where v is characteristic velocity, U is Newtonian gravitational potential, and c is speed of light.

- **Order counting:** Newtonian order $O(\epsilon)$, 1PN order $O(\epsilon^2)$
- **Gauge condition:** Harmonic/Lorenz gauge $\partial_\mu \bar{h}^{\mu\nu} = 0$, where $\bar{h}^{\mu\nu} = h^{\mu\nu} - \frac{1}{2}\eta^{\mu\nu}h$
- **Field equations:** $\bar{h}_{\mu\nu} = -2\kappa T_{\mu\nu}$ to consistent PN order

15.2 Matter variables and conservation laws

Define the matter variables in the fluid description:

- **Rest mass density:** $\rho(\mathbf{x}, t)$
- **Coordinate 3-velocity:** $v^i = dx^i/dt$
- **Pressure:** $p(\mathbf{x}, t)$
- **Internal energy density:** $\Pi(\mathbf{x}, t)$ (per unit mass)
- **Stress tensor:** $\sigma^{ij} = \rho v^i v^j + p\delta^{ij}$

The continuity and Euler equations at Newtonian order:

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0 \tag{42}$$

$$\rho (\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v}) = -\nabla p + \rho \nabla U \tag{43}$$

15.3 Potential definitions at 1PN order

The gravitational potentials satisfy Poisson-type equations with specific sources:

$$\nabla^2 U = -4\pi G \rho$$

$$\nabla^2 V_i = -4\pi G \rho v_i$$

$$\nabla^2 \Phi_1 = -4\pi G \rho v^2$$

$$\nabla^2 \Phi_2 = -4\pi G \rho U$$

$$\nabla^2 \Phi_3 = -4\pi G \rho \Pi$$

$$\nabla^2 \Phi_4 = -4\pi G p$$

Their integral representations:

$$\begin{aligned}
U(\mathbf{x}, t) &= G \int \frac{\rho(\mathbf{x}', t)}{|\mathbf{x} - \mathbf{x}'|} d^3 x' \\
V_i(\mathbf{x}, t) &= G \int \frac{\rho(\mathbf{x}', t) v_i(\mathbf{x}', t)}{|\mathbf{x} - \mathbf{x}'|} d^3 x' \\
\Phi_1(\mathbf{x}, t) &= G \int \frac{\rho(\mathbf{x}', t) v^2(\mathbf{x}', t)}{|\mathbf{x} - \mathbf{x}'|} d^3 x' \\
\Phi_2(\mathbf{x}, t) &= G \int \frac{\rho(\mathbf{x}', t) U(\mathbf{x}', t)}{|\mathbf{x} - \mathbf{x}'|} d^3 x' \\
\Phi_3(\mathbf{x}, t) &= G \int \frac{\rho(\mathbf{x}', t) \Pi(\mathbf{x}', t)}{|\mathbf{x} - \mathbf{x}'|} d^3 x' \\
\Phi_4(\mathbf{x}, t) &= G \int \frac{p(\mathbf{x}', t)}{|\mathbf{x} - \mathbf{x}'|} d^3 x'
\end{aligned}$$

The composite 1PN potential: $\Psi := \Phi_1 + \Phi_2 + \Phi_3 + 3\Phi_4$

15.4 Metric tensor at 1PN order

Since TEGR is equivalent to GR at the level of field equations, the 1PN metric matches the standard GR result:

$$\begin{aligned}
g_{00} &= -1 + \frac{2U}{c^2} - \frac{2}{c^4} (U^2 - 2\Psi) + O(c^{-6}) \\
g_{0i} &= -\frac{4}{c^3} V_i + O(c^{-5}) \\
g_{ij} &= \delta_{ij} \left(1 + \frac{2U}{c^2} \right) + O(c^{-4})
\end{aligned} \tag{44}$$

This satisfies the harmonic gauge condition $\partial_\mu (\sqrt{-g} g^{\mu\nu}) = 0$ to 1PN order.

15.5 Gravitoelectromagnetism at 1PN order

Using the GEM correspondence:

$$\begin{aligned}
\Phi_g &= -\frac{c^2}{2} h_{00} = -\frac{c^2}{2} (g_{00} + 1) \\
\mathbf{A}_g &= \frac{c^2}{2} (h_{01}, h_{02}, h_{03}) = \frac{c^2}{2} g_{0i}
\end{aligned}$$

Substituting the 1PN metric:

$$\begin{aligned}
\Phi_g &= U - \frac{1}{c^2} (U^2 - 2\Psi) + O(c^{-4}) \\
\mathbf{A}_g &= -\frac{2}{c} \mathbf{V} + O(c^{-3})
\end{aligned} \tag{45}$$

The GEM fields then inherit 1PN corrections:

$$\begin{aligned}\mathbf{E}_g &= -\nabla\Phi_g - \partial_t\mathbf{A}_g \\ \mathbf{B}_g &= \nabla \times \mathbf{A}_g\end{aligned}$$

15.6 Classical tests at 1PN order

Perihelion precession

For a test mass in Schwarzschild geometry (mass M), the pericenter advance per orbit:

$$\Delta\omega = \frac{6\pi GM}{a(1-e^2)c^2} + O(c^{-4}) \quad (46)$$

where a is semi-major axis, e is eccentricity.

Light deflection

For a light ray with impact parameter b :

$$\alpha = \frac{4GM}{bc^2} + O(c^{-4}) \quad (47)$$

Shapiro time delay

The gravitational time delay for a signal from r_1 to r_2 with closest approach b :

$$\Delta t_{\text{Shapiro}} = \frac{2GM}{c^3} \ln \left(\frac{r_1 + r_2 + D}{r_1 + r_2 - D} \right) + O(c^{-5}) \quad (48)$$

where $D = \sqrt{r_1^2 + r_2^2 - b^2}$.

15.7 Two-body dynamics at 1PN order

For a binary system with masses m_1, m_2 , define:

- Total mass: $M = m_1 + m_2$
- Reduced mass: $\mu = m_1 m_2 / M$
- Symmetric mass ratio: $\eta = \mu / M$
- Relative position: $\mathbf{r} = \mathbf{x}_1 - \mathbf{x}_2$, $r = |\mathbf{r}|$
- Relative velocity: $\mathbf{v} = d\mathbf{r}/dt$
- Radial velocity: $\dot{r} = \mathbf{n} \cdot \mathbf{v}$, where $\mathbf{n} = \mathbf{r}/r$

The relative acceleration in center-of-mass frame:

$$\mathbf{a} = -\frac{GM}{r^2}\mathbf{n} + \frac{GM}{c^2 r^2}\left[\mathbf{n}\left\{(4+2\eta)\frac{GM}{r} - (1+3\eta)v^2 + \frac{3}{2}\eta\dot{r}^2\right\} + (4-2\eta)\dot{r}\mathbf{v}\right] + O(c^{-4}) \quad (49)$$

15.8 Bimetric extension (Yukawa corrections)

If a massive graviton component exists with mass μ_g , the potential splits as:

$$\begin{aligned} U(\mathbf{x}) &= U_0(\mathbf{x}) + U_m(\mathbf{x}) \\ U_0(\mathbf{x}) &= G \int \frac{\rho(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x' \\ U_m(\mathbf{x}) &= G \sin \vartheta \int \frac{\rho(\mathbf{x}') e^{-\mu_g |\mathbf{x} - \mathbf{x}'|}}{|\mathbf{x} - \mathbf{x}'|} d^3x' \end{aligned}$$

where ϑ is the mixing angle. Solar System constraints require:

- Either $\mu_g^{-1} \gg 1$ AU (long range)
- Or $\sin \vartheta \ll 1$ (small mixing)

15.9 Dimensional analysis

- $[U] = [\Phi_A] = \text{m}^2 \text{s}^{-2}$ (gravitational potential)
- $[V_i] = \text{m}^3 \text{s}^{-3}$ (vector potential)
- $[GM/(ac^2)] = 1$ (dimensionless perihelion factor)
- $[GM/(bc^2)] = 1$ (dimensionless deflection)
- $[GM/c^3] = \text{s}$ (Shapiro delay scale)

All terms in the 1PN metric are properly dimensionless, and the classical test predictions have correct physical dimensions.

$$\begin{aligned} \text{1PN framework: } g_{\mu\nu} &= \eta_{\mu\nu} + h_{\mu\nu}^{(1)} + h_{\mu\nu}^{(2)} + O(h^3) \\ \text{TEGR = GR to 1PN: same dynamics, different geometric interpretation} \end{aligned} \quad (50)$$

16 Gravitational waves

Vacuum wave equation and TT gauge

Linearized, Lorenz gauge: $\bar{h}_{\mu\nu} = 0$ in vacuum. Project to TT:

$$h_{0\mu}^{\text{TT}} = 0, \quad \partial_i h_{ij}^{\text{TT}} = 0, \quad h_{ii}^{\text{TT}} = 0.$$

Plane wave along $\hat{\mathbf{n}}$:

$$h_{ij}^{\text{TT}}(x) = \Re \left[(A_+ e_{ij}^+ + A_\times e_{ij}^\times) e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} \right], \quad \omega = c k, \quad e_{ij}^{+, \times} e_{ij}^{+, \times} = 2.$$

$$h_{ij}^{\text{TT}} = 0, \quad \partial_i h_{ij}^{\text{TT}} = 0, \quad h_{ii}^{\text{TT}} = 0 \quad (51)$$

GEM fields for a plane GW

For propagation along $\hat{\mathbf{n}}$,

$$\mathbf{B}_g = \frac{1}{c} \hat{\mathbf{n}} \times \mathbf{E}_g, \quad \|\mathbf{E}_g\| = \|\mathbf{B}_g\| c, \quad \omega = ck.$$

Effective energy–momentum (Isaacson)

Averaging over several cycles,

$$\langle t_{\mu\nu}^{\text{GW}} \rangle = \frac{1}{32\pi G} \langle \partial_\mu h_{ij}^{\text{TT}} \partial_\nu h_{ij}^{\text{TT}} \rangle \quad (52)$$

Energy flux along $\hat{\mathbf{n}}$:

$$\langle S \rangle = \frac{c^3}{32\pi G} \langle \dot{h}_{ij}^{\text{TT}} \dot{h}_{ij}^{\text{TT}} \rangle \hat{\mathbf{n}} \quad (53)$$

Quadrupole formulas (far zone)

Reduced mass quadrupole $I_{ij} = \int \rho (x_i x_j - \frac{1}{3} \delta_{ij} r^2) d^3x$. At distance R :

$$h_{ij}^{\text{TT}}(t, \mathbf{x}) = \frac{2G}{c^4 R} \ddot{I}_{ij}^{\text{TT}} \left(t - \frac{R}{c} \right) \quad (54)$$

Total radiated power:

$$P = \frac{G}{5c^5} \langle \ddot{I}_{ij} \ddot{I}_{ij} \rangle \quad (55)$$

Binary inspiral (circular, leading order) Chirp mass $M_c = (m_1 m_2)^{3/5} / (m_1 + m_2)^{1/5}$, GW frequency $f = \omega / \pi$:

$$h(f) \simeq \frac{4}{R} \frac{G^{5/3}}{c^4} M_c^{5/3} (\pi f)^{2/3} \quad (56)$$

Memory

Permanent offset after a burst: linear memory from $\Delta \dot{I}_{ij}$, nonlinear (Christodoulou) from GW self-flux. Schematic far-zone jump:

$$\Delta h_{ij}^{\text{TT}} \sim \frac{4G}{c^4 R} \Delta \int dt \mathcal{F}_{ij}^{(\text{source+GW})} \quad (\text{tracefree, TT}).$$

Massive (bimetric) radiative sector

Massive eigenmode obeys $(-\mu^2)\hat{h}_{\mu\nu}^{(m)} = 0$, with five polarizations at linear order. Dispersion and group speed:

$$\omega^2 = c^2(k^2 + \mu^2), \quad v_g = \frac{c^2 k}{\omega} < c \quad (57)$$

Far zone strain acquires $e^{-\mu R}$ suppression and phase lag; emission is cut off for source frequencies $\omega < \mu c$.

Units

$[h_{ij}] = 1$, $[\langle S \rangle] = \text{W m}^{-2}$, $[P] = \text{W}$, $[I_{ij}] = \text{kg m}^2$, $[\mu] = \text{m}^{-1}$. Equations (52)–(55) are dimensionally consistent.

17 Gauge/tetrad catalog

Principles

- **Covariant TEGR:** any tetrad reproducing $g_{\mu\nu}$ is admissible if accompanied by the inertial spin connection $\omega^a_{b\mu}$ (pure gauge).
- **Pure-tetrad gauge** ($\omega = 0$): use only “good” tetrads that make the antisymmetric field equation vanish; typically non-diagonal or locally rotated frames.
- **Rule of thumb:** in curvilinear coordinates, include the flat-space inertial ω of the coordinate frame or absorb it by a local Lorentz rotation of the tetrad.

Minkowski, Cartesian (trivial)

Metric $\eta_{\mu\nu}$. Choose $e^a_{\mu} = \delta^a_{\mu}$, $\omega^a_{b\mu} = 0$ (both covariant and pure-tetrad OK).

Minkowski, spherical coordinates

Coordinate line element $ds^2 = -dt^2 + dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)$. Use the diagonal orthonormal tetrad:

$$e^a_{\mu} = \text{diag}(1, 1, r, r \sin \theta).$$

In covariant TEGR supply the *inertial* nonzero components (flat space):

$$\omega^{\hat{r}}_{\hat{\theta}\theta} = -1, \quad \omega^{\hat{r}}_{\hat{\phi}\phi} = -\sin \theta, \quad \omega^{\hat{\theta}}_{\hat{\phi}\phi} = -\cos \theta. \quad (58)$$

In pure-tetrad gauge, instead rotate the frame by the standard spherical triad $\Lambda(\theta, \phi)$ so that $\omega = 0$.

Schwarzschild (static, spherical)

$ds^2 = -A dt^2 + A^{-1} dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)$, $A(r) = 1 - \frac{2GM}{c^2 r}$. Diagonal orthonormal tetrad:

$$e^a{}_\mu = \text{diag}(\sqrt{A}, A^{-1/2}, r, r \sin \theta).$$

Covariant TEGR: use the *same* inertial ω as in (58). Pure-tetrad (good) choice: apply the spherical rotation $\Lambda(\theta, \phi)$ to the spatial legs:

$$e'^a{}_\mu = \Lambda^a{}_b(\theta, \phi) e^b{}_\mu, \quad \omega' = 0 \quad (\text{good pure tetrad}). \quad (59)$$

Stationary axisymmetric metrics (Kerr form, generic recipe)

Write the line element in t - ϕ block form

$$ds^2 = -N^2 dt^2 + g_{rr} dr^2 + g_{\theta\theta} d\theta^2 + g_{\phi\phi} (d\phi + \beta dt)^2,$$

where $N^2 := -(g_{tt} - g_{t\phi}^2/g_{\phi\phi})$ and $\beta := g_{t\phi}/g_{\phi\phi}$. An orthonormal coframe (1-forms) is then

$$\theta^{(0)} = N dt, \quad \theta^{(1)} = \sqrt{g_{rr}} dr, \quad \theta^{(2)} = \sqrt{g_{\theta\theta}} d\theta, \quad \theta^{(3)} = \sqrt{g_{\phi\phi}} (d\phi + \beta dt).$$

Use $e^a{}_\mu$ defined by $\theta^{(a)} = e^{(a)}{}_\mu dx^\mu$ and include the inertial ω determined by the r, θ -dependent triad.

$$\text{Stationary orthonormal tetrad: } e^{(0)}{}_t = N, \quad e^{(3)}{}_\phi = \sqrt{g_{\phi\phi}}, \quad e^{(3)}{}_t = \sqrt{g_{\phi\phi}} \beta, \quad \text{others diagonal.} \quad (60)$$

FRW cosmology

Flat FRW: $ds^2 = -dt^2 + a^2(t) \delta_{ij} dx^i dx^j$. Good diagonal tetrad works in both covariant and pure-tetrad forms:

$$e^a{}_\mu = \text{diag}(1, a(t), a(t), a(t)), \quad \omega^a{}_{b\mu} = 0.$$

Curved FRW in spherical comoving coordinates uses $e^a{}_\mu = \text{diag}(1, a/\sqrt{1-kr^2}, ar, ar \sin \theta)$ with inertial ω as in (58).

Weak-field (harmonic/Lorenz teleparallel gauge)

$e^a{}_\mu = \delta^a{}_\mu + \frac{1}{2} h^a{}_\mu$, $\partial_\nu (e S^{\mu\nu}{}_\lambda) = 0$. Map to GEM via Φ_g, A_g (sections above); $\omega = 0$ is consistent to linear order in Cartesian charts.

Rotating frames in flat spacetime (cylindrical)

In coordinates (t, r, ϕ, z) with angular velocity Ω , $g_{t\phi} = \Omega r^2$. Use the stationary recipe (60) with $N = \sqrt{1 - \Omega^2 r^2/c^2}$ and $\beta = \Omega$.

Checklist for a “good” choice

1. Reproduce $g_{\mu\nu} = \eta_{ab}e^a_\mu e^b_\nu$ exactly.
2. If using curvilinear coordinates, add flat-space inertial ω (covariant) or rotate the tetrad to absorb it (pure-tetrad).
3. Verify antisymmetric field eq. vanishes (pure-tetrad) or is identically satisfied (covariant).
4. Confirm correct Newtonian limit and Lense–Thirring terms in weak fields.
5. Check units and signatures; ensure $e > 0$ and correct orientation.

18 3+1 Hamiltonian

Split and time gauge

Metric split:

$$ds^2 = -N^2 dt^2 + \gamma_{ij}(dx^i + N^i dt)(dx^j + N^j dt).$$

Time gauge tetrad:

$$e^{(0)}_0 = N, \quad e^{(0)}_i = 0, \quad e^{(a)}_0 = N^j E^{(a)}_j, \quad e^{(a)}_i = E^{(a)}_i,$$

$$\gamma_{ij} = \delta_{ab} E^{(a)}_i E^{(b)}_j, \quad e = N\sqrt{\gamma}.$$

Canonical pair and spatial torsion

Spatial torsion of the triad:

$$T^{(a)}_{ij} := \partial_i E^{(a)}_j - \partial_j E^{(a)}_i, \quad T_i := T^{(a)}_{ai} = E_{(a)}^j T^{(a)}_{ji}.$$

Canonical momentum (from TEGR Lagrangian):

$$\Pi_{(a)}^i := \frac{\partial \mathcal{L}}{\partial(\partial_0 E^{(a)}_i)} = \frac{e}{\kappa} S_{(a)}^{0i} \quad (61)$$

Constraints

Primary: $\Pi_{(a)}^0 \approx 0$. SO(3) (rotations) on the slice:

$$\mathcal{G}_{ab} := \Pi_{[a}^i E_{b]i} \approx 0 \quad (62)$$

Vector (spatial diffeomorphisms):

$$\mathcal{H}_i := \Pi_{(a)}^j \partial_i E^{(a)}_j - \partial_j (\Pi_{(a)}^j) E^{(a)}_i \approx 0 \quad (63)$$

Scalar (Hamiltonian) in teleparallel form:

$$\mathcal{T} := \frac{1}{4} T_{(a)ij} T^{(a)ij} + \frac{1}{2} T_{(a)ij} T^{ji(a)} - T_i T^i,$$

$$\mathcal{H} := \frac{\kappa}{e} \left(\Pi_{(a)}^i \Pi_{(a)i} - \frac{1}{2} (\Pi_{(a)}^i E_{(a)i})^2 \right) - e \mathcal{T} - \partial_i (2e T^i) \approx 0 \quad (64)$$

Hamiltonian and boundary term

$$H = \int d^3x \left(N \mathcal{H} + N^i \mathcal{H}_i + \frac{1}{2} \lambda^{ab} \mathcal{G}_{ab} + \lambda^{(a)} \Pi_{(a)}^{(0)} \right) + B \quad (65)$$

Quasilocal boundary:

$$B = \frac{2}{\kappa} \oint_{\partial\Sigma} dS_i e S_{(0)}^{0i} \quad (66)$$

DOF count

Variables: $E^{(a)}_i$ (9) and $\Pi_{(a)}^i$ (9). First-class constraints: $\mathcal{H}$ (1), $\mathcal{H}_i$ (3), $\mathcal{G}_{ab}$ (3). Result: $18 - 2 \times 7 = 4$ phase-space DOF $\Rightarrow$ 2 configuration DOF (GR equivalence).

Units

$$[E^{(a)}_i] = 1, [\Pi_{(a)}^i] = \text{m}^{-1}/\kappa, [\mathcal{H}] = \text{m}^{-3}/\kappa, [B] = 1/\kappa.$$

19 Nonlinear HGEM

Second order in amplitude

Let $h_{\mu\nu} = \epsilon h_{\mu\nu}^{(1)} + \epsilon^2 h_{\mu\nu}^{(2)} + O(\epsilon^3)$ in Lorenz gauge. Then

$$\begin{aligned} \bar{h}_{\mu\nu}^{(1)} &= -2\kappa T_{\mu\nu}, \\ \bar{h}_{\mu\nu}^{(2)} &= -2\kappa T_{\mu\nu}^{(2)} + \mathcal{N}_{\mu\nu}[h^{(1)}, h^{(1)}], \end{aligned}$$

where $\mathcal{N}_{\mu\nu}$ collects quadratic self-interactions (from $R_{\mu\nu}^{(2)}$ or, equivalently, from a pseudotensor source). Projecting to GEM variables gives

$$\Phi_g^{(2)} = 4\pi G \rho^{(2)} + S_\Phi[\Phi_g^{(1)}, A_g^{(1)}], \quad A_g^{(2)} = \frac{4\pi G}{c^2} \mathbf{J}^{(2)} + S_A[\Phi_g^{(1)}, A_g^{(1)}],$$

with $S_\Phi, S_A = O(\epsilon^2)$ built from quadratic GEM fields.

Effective quadratic source (Isaacson form)

Define instantaneous GW effective stress (no cycle average) by

$$t_{\mu\nu}^{\text{GW}} := \frac{1}{32\pi G} \partial_\mu h_{ij}^{\text{TT}} \partial_\nu h_{ij}^{\text{TT}} \quad \Rightarrow \quad \mathcal{N}_{\mu\nu} \hat{=} 16\pi G \left(t_{\mu\nu}^{\text{GW}} - \frac{1}{2} \eta_{\mu\nu} t^{\text{GW}} \right),$$

which yields concrete S_Φ, S_A after the GEM map.

Modal expansion and normalization

In a cavity Ω , expand time-harmonic first-order fields as

$$\Phi_g^{(1)} = \sum_a a_a(t) \phi_a(\mathbf{x}) e^{-i\omega_a t} + c.c., \quad \mathbf{A}_g^{(1)} = \sum_a a_a(t) \mathbf{u}_a(\mathbf{x}) e^{-i\omega_a t} + c.c.$$

with $(\nabla^2 + k_a^2)\{\phi_a, \mathbf{u}_a\} = 0$, $k_a = \omega_a/c$, Lorenz constraint $\nabla \cdot \mathbf{u}_a + i(\omega_a/c^2)\phi_a = 0$. Normalize modal energies by

$$U_a = \frac{1}{8\pi G} \int_{\Omega} \left(\frac{\|\mathbf{E}_a\|^2}{c^2} + \|\mathbf{B}_a\|^2 \right) dV, \quad \mathbf{E}_a := -\nabla \phi_a, \quad \mathbf{B}_a := \nabla \times \mathbf{u}_a \quad (67)$$

Three-wave mixing (projection)

Project $\Phi_g^{(2)}$ and $\mathbf{A}_g^{(2)}$ on mode a . Keeping near-resonant triads (a, b, c) with $\omega_a \approx \omega_b \pm \omega_c$,

$$\dot{a}_a + (\gamma_a + i\Delta\omega_a) a_a = i \sum_{b,c} \left(\chi_{abc}^{(+)} a_b a_c e^{-i\Delta_{abc}^{(+)} t} + \chi_{abc}^{(-)} a_b a_c^* e^{-i\Delta_{abc}^{(-)} t} \right) \quad (68)$$

with detunings $\Delta_{abc}^{(+)} = \omega_b + \omega_c - \omega_a$ and $\Delta_{abc}^{(-)} = \omega_b - \omega_c - \omega_a$, linear loss γ_a , and nonlinear coefficients

$$\chi_{abc}^{(\pm)} = \frac{1}{2U_a} \int_{\Omega} \mathcal{Q}_{abc}^{(\pm)}(\phi, \mathbf{u}) dV,$$

where $\mathcal{Q}_{abc}^{(\pm)}$ are quadratic overlap densities derived from $\mathbf{S}_{\Phi}, \mathbf{S}_A$ (built from products like $\mathbf{E}_b \cdot \mathbf{E}_c / c^2, \mathbf{B}_b \cdot \mathbf{B}_c$, and mixed terms satisfying the Lorenz constraint).

Selection rules and Manley–Rowe

Nonzero χ_{abc} require (i) frequency matching (small detuning), (ii) spatial overlap $\int \mathcal{Q}_{abc} \neq 0$ dictated by cavity symmetry. Energy exchange obeys Manley–Rowe-type invariants for exact resonance:

$$\frac{d}{dt} \left(\frac{|a_a|^2}{\omega_a} \pm \frac{|a_b|^2}{\omega_b} \pm \frac{|a_c|^2}{\omega_c} \right) = 0,$$

with signs matching the + or – process in (68).

Self-phase modulation and detuning

The second-order self-energy shifts $\omega_a \rightarrow \omega_a + \Delta\omega_a$ with

$$\Delta\omega_a = \frac{1}{2\omega_a U_a} \int_{\Omega} \left(\frac{\|\mathbf{E}^{(1)}\|^2}{c^2} + \|\mathbf{B}^{(1)}\|^2 \right)_a^{(\text{proj})} dV \sim O(h^2) \quad (69)$$

where “proj” denotes projection onto mode a by the Lorenz-weighted inner product.

Parametric instability threshold

For a strong pump p driving $b \leftrightarrow c$, the small-signal gain $g \approx |\chi_{pbc}| |a_p| - \frac{\gamma_b + \gamma_c}{2}$. Instability if $|\chi_{pbc}| |a_p| > \frac{\gamma_b + \gamma_c}{2}$.

Cubic envelope (self-Kerr analogue)

Keeping only a_a and its self-coupling yields

$$\dot{a}_a + (\gamma_a + i\Delta\omega_{0a})a_a = i\xi_a |a_a|^2 a_a, \quad \xi_a = \frac{1}{2U_a} \int_{\Omega} \mathcal{Q}_{aaaa} dV \quad (70)$$

with $\Delta\omega_{0a}$ the linear detuning and $\xi_a = O(G)$ extremely small.

Units

$[U_a] = J$, $[\chi_{abc}] = s^{-1}$, $[\xi_a] = s^{-1}$, $[\Delta\omega_a] = s^{-1}$. Amplitudes a_a are dimensionless under the normalization (67).

Remarks

Gravitational nonlinearity is minute: for strain $h \sim 10^{-n}$, shifts scale as $O(h^2)$. Equations (68)–(70) capture mixing, self-detuning, and thresholds needed for HGEM cavity studies.

20 Stability and IBVP

IBVP for GEM potentials (Lorenz gauge)

Domain $\Omega \subset \mathbb{R}^3$ with boundary $\Gamma = \Gamma_s \cup \Gamma_v \cup \Gamma_0$. Unknowns: $\Phi_g, \mathbf{A}_g$ with

$$\Phi_g = 4\pi G \rho, \quad \mathbf{A}_g = \frac{4\pi G}{c^2} \mathbf{J}, \quad C := \nabla \cdot \mathbf{A}_g + \frac{1}{c^2} \partial_t \Phi_g = 0.$$

Boundary data:

$$\begin{aligned} \Gamma_s : \quad & (\partial_n - \frac{1}{c} \partial_t) \Phi_g + \eta_s \Phi_g = g_s, \\ \Gamma_v : \quad & \mathbf{n} \times \mathbf{B}_g - \frac{1}{c} \mathbf{n} \times (\mathbf{n} \times \mathbf{E}_g) + \eta_v \mathbf{n} \times \mathbf{A}_g = \mathbf{g}_v, \\ \Gamma_0 : \quad & \partial_n \Phi_g = 0, \quad \mathbf{n} \times \mathbf{A}_g = \mathbf{0}, \end{aligned}$$

with $\mathbf{E}_g = -\nabla \Phi_g - \partial_t \mathbf{A}_g$, $\mathbf{B}_g = \nabla \times \mathbf{A}_g$. Initial data $(\Phi_0, \Phi_1, \mathbf{A}_0, \mathbf{A}_1)$ satisfy $C|_{t=0} = 0$ and boundary compatibility.

Energy identity and uniqueness

Define

$$E(t) := \frac{1}{8\pi G} \int_{\Omega} \left(\frac{\|\mathbf{E}_g\|^2}{c^2} + \|\mathbf{B}_g\|^2 \right) dV.$$

For smooth solutions one obtains

$$\frac{dE}{dt} + \int_{\Gamma} \mathbf{S}_g \cdot \mathbf{n} dA = -\frac{1}{c^2} \int_{\Omega} \mathbf{J} \cdot \mathbf{E}_g dV, \quad \mathbf{S}_g := -\frac{1}{4\pi G} \mathbf{E}_g \times \mathbf{B}_g \quad (71)$$

With $\eta_{s,v} \geq 0$, $g_s = \mathbf{g}_v = \rho = \mathbf{J} = 0$, (71) gives $E(t) \leq E(0)$, hence uniqueness by $E(0) = 0 \Rightarrow E(t) \equiv 0$.

Constraint propagation and damping

Generalized Lorenz gauge-fix: add $-\frac{\alpha}{2}C^2$ to the Lagrangian ($\alpha > 0$). The modified system yields

$$C + \alpha \partial_t C = \frac{4\pi G}{c^2} (\partial_t \rho + \nabla \cdot \mathbf{J}) \quad (72)$$

So if the continuity equation holds, C obeys a damped wave and $C \rightarrow 0$ as $t \rightarrow \infty$.

Well-posedness (semigroup form)

Let $\mathcal{A}$ be the Maxwell-like generator on $(\mathbf{E}_g, \mathbf{B}_g)$ with impedance data on $\Gamma_{s,v}$ and perfect conductor on Γ_0 . Then $\mathcal{A}$ is maximal dissipative in the energy norm induced by $E(t)$.

$$\|U(t)\|_E \leq \|U(0)\|_E + \int_0^t (\|f(\tau)\| + \|g(\tau)\|) d\tau, \quad U := (\mathbf{E}_g, \mathbf{B}_g) \quad (73)$$

implying existence, uniqueness, and continuous dependence on data.

Absorbing and PML layers

On an exterior shell Ω_{PML} use split-field PML with profiles $\sigma_{x,y,z} \geq 0$:

$$\partial_t \mathbf{E}_g + \Sigma \mathbf{E}_g - c^2 \nabla \times \mathbf{B}_g = \dots, \quad \partial_t \mathbf{B}_g + \Sigma \mathbf{B}_g + \nabla \times \mathbf{E}_g = \dots,$$

where $\Sigma = \text{diag}(\sigma_x, \sigma_y, \sigma_z)$. Choose σ so that the PML contribution adds a nonnegative damping term to dE/dt .

Discrete stability (FDTD/CFL)

Yee-type leapfrog on a Cartesian grid with spacing $\Delta x, \Delta y, \Delta z$ is stable if

$$\Delta t \leq \frac{1}{c} \frac{1}{\sqrt{\Delta x^{-2} + \Delta y^{-2} + \Delta z^{-2}}} \equiv \Delta t_{\max} \quad (74)$$

Impedance or PML boundaries do not reduce (74) for explicit schemes under standard implementations.

Compatibility conditions

At $t = 0$:

$$C(\Phi_0, \mathbf{A}_0) = 0, \quad (\partial_n - \frac{1}{c} \partial_t) \Phi_g|_{\Gamma_{s,0}} = g_s(0), \quad \mathbf{n} \times \mathbf{B}_g|_{\Gamma_{v,0}} - \frac{1}{c} \mathbf{n} \times (\mathbf{n} \times \mathbf{E}_g)|_0 = \mathbf{g}_v(0).$$

Sources satisfy $\partial_t \rho + \nabla \cdot \mathbf{J} = 0$.

Units

$[E] = \text{J}$, $[\mathbf{S}_g] = \text{W m}^{-2}$, $[\alpha] = \text{s}^{-1}$, consistent with (71)–(74).

21 Implementation details

Code layout

- `core/`: field types, units, geometry, boundary tags.
- `pde/`: GEM operators, Lorenz constraint, sources.
- `time/`: explicit FDTD (Yee), RK(4), IMEX(2).
- `freq/`: FEM Helmholtz (mixed + penalty).
- `lin/`: Krylov (CG, GMRES), AMG/ILU preconditioners.
- `io/`: mesh import (Gmsh), VTK output, JSON config.

Units and scaling

SI internally. Optional nondimensionalization by L_0 , $T_0 = L_0/c$, $E_0 = \Phi_0/L_0$. Scale sources: $\rho' = \rho/(\rho_0)$, $\mathbf{J}' = \mathbf{J}/(\rho_0 c)$.

Time domain: staggered-grid FDTD (Lorenz gauge)

Yee grid: $\mathbf{E}_g$ and $\mathbf{B}_g$ staggered, Φ_g and $\mathbf{A}_g$ collocated. Gauge update by penalty α or auxiliary C -field.

$$\begin{aligned}
 \mathbf{B}^{n+\frac{1}{2}} &= \mathbf{B}^{n-\frac{1}{2}} - \Delta t \nabla \times \mathbf{E}^n, \\
 \mathbf{E}^{n+1} &= \mathbf{E}^n - \Delta t \left(\nabla \times \mathbf{B}^{n+\frac{1}{2}} + \frac{4\pi G}{c^2} \mathbf{J}^{n+\frac{1}{2}} \right), \\
 \Phi^{n+1} &= \Phi^n - \alpha \Delta t C^n - \Delta t c^2 \nabla \cdot \mathbf{A}^{n+\frac{1}{2}} + \frac{4\pi G}{|\Omega|} \int_{\Omega} \rho^n dV, \\
 \mathbf{A}^{n+\frac{1}{2}} &= \mathbf{A}^{n-\frac{1}{2}} - \Delta t (\mathbf{E}^n + \nabla \Phi^n), \quad C^n = \nabla \cdot \mathbf{A}^{n-\frac{1}{2}} + \frac{1}{c^2} \partial_t \Phi^n
 \end{aligned} \tag{75}$$

CFL: $\Delta t \leq \Delta t_{\max}$ from (74). Impedance/PML on outer layers.

Frequency domain: mixed FEM (Helmholtz + Lorenz)

Spaces: H^1 for Φ, λ , Nédélec $H(\text{curl})$ for $\mathbf{A}$. Linear system (after assembly and boundary terms):

$$\begin{bmatrix} K_{\Phi} - k^2 M_{\Phi} & 0 & i \frac{\omega}{c^2} M_{\Phi \lambda} \\ 0 & K_{\text{curl}} - k^2 M_A & D^{\top} \\ i \frac{\omega}{c^2} M_{\lambda \Phi} & D & 0 \end{bmatrix} \begin{bmatrix} \Phi \\ \mathbf{A} \\ \lambda \end{bmatrix} = \begin{bmatrix} 4\pi G b_{\rho} \\ \frac{4\pi G}{c^2} b_J \\ 0 \end{bmatrix} \tag{76}$$

Preconditioning: block-diagonal $(K_{\Phi} + k^2 M_{\Phi})$, $(K_{\text{curl}} + k^2 M_A)$, and mass for λ ; AMG on H^1 blocks, Hiptmair–Xu for $H(\text{curl})$.

Boundary conditions

Sommerfeld: $(\partial_n - ik)\Phi = 0$ on Γ_s . Silver–Müller: $\mathbf{n} \times (\nabla \times \mathbf{A}) - ik \mathbf{n} \times (\mathbf{n} \times \mathbf{A}) = \mathbf{0}$ on Γ_v . Perfect conductor analogue: $\partial_n \Phi = 0$, $\mathbf{n} \times \mathbf{A} = \mathbf{0}$ on Γ_0 .

Source injection

Soft source in time-domain via Gaussian-modulated sine in $\rho, \mathbf{J}$ with continuity enforced by $\partial_t \rho + \nabla \cdot \mathbf{J} = 0$. In frequency-domain, use projected right-hand sides b_ρ, b_J consistent with discrete divergence.

Solvers and tolerances

- Time-domain: monitor energy $E(t)$ and Lorenz residual $\|C\|_2$; enforce $\|C\|_2/\|\Phi\|_2 \leq 10^{-6}$.
- Frequency-domain: GMRES with restart 50, rel. residual 10^{-8} ; shift-and-invert for eigenmodes.

Convergence and mesh/time-step studies

Refine h and Δt with manufactured solutions: $\Phi^* = \sin(\pi x) \sin(\pi y) \sin(\pi z)$, $\mathbf{A}^* = \nabla \times (\Phi^* \hat{\mathbf{z}})/\pi$. Expect $\|\cdot\|_2$ rates: second order in time for leapfrog, first/second order in space depending on stencil; FEM $O(h^p)$ with polynomial degree p .

Verification set

- Static point mass (20); measure $\|\Phi - \Phi_{\text{ana}}\|_{L^2}$.
- Dipole frame-dragging (21); check $\mathbf{B}_g$ decay $\propto r^{-3}$.
- Cavity spectra (22) and (23); compare $k_{\ell n}$.
- Retarded pulse vs analytic Green's function (8).

Stop criteria:
$$\begin{cases} \text{Time: } \Delta E/E_0 < 10^{-8} \wedge \|C\|_2 < 10^{-6} \\ \text{Freq: } \|r\|_2/\|b\|_2 < 10^{-8} \text{ or } |\Delta k|/k < 10^{-8} \end{cases}$$

(77)